Girmay Meresu

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Professional Summary

Third-year Software Engineering student specializing in AI systems architecture, local data pipelines, and intelligent application design. Hands-on experience developing local RAG pipelines, relational database systems, and robust backend utilities using Python and C++. Blends strong technical programming foundations with practical digital design and multimedia tooling expertise.

Education

Addis Ababa Science and Technology University (AASTU) — Addis Ababa, Ethiopia

- **Bachelor of Science in Software Engineering** (Expected Graduation: Present — 2023/2024)
- **Current Standing:** 3rd Year, Semester I
- **Key Coursework:** Data Structures & Algorithms, Object Oriented Programming, Internet Programming I, Systems Analysis and Management, Computer Organization and Architecture.

Technical Skills

- **Programming Languages:** Python, C++, Java (Basic)
- **AI & Data Pipelines:** Sentence Transformers, Local LLMs (Ollama / Llama 3.2), Vector Embeddings, Data Ingestion & Chunking
- **Databases & Vector Stores:** Qdrant Vector DB, ChromaDB, MySQL (Relational Databases, SQL Procedures, ER Diagrams)
- **Developer Tools:** Git, GitHub, Docker, Docker Compose, VS Code, Chrome DevTools
- **Creative & Multimedia Tools:** Adobe Photoshop, Adobe Premiere Pro, Adobe Illustrator, InDesign, Canva, CapCut, Fast Typing

Projects

Non-Agentic RAG Pipeline | Python, Qdrant, Sentence-Transformers, Ollama (Llama 3.2:3B)

- Engineered a local Retrieval-Augmented Generation (RAG) pipeline supporting full data ingestion, document chunking, and local vector embedding.
- Integrated `sentence-transformers` for local text encoding and indexed vectors within a standalone Qdrant vector database container.
- Configured local response generation using Llama 3.2:3B via Ollama to query domain-specific documents securely offline.

Python Web Scraper | Python, Requests, BeautifulSoup

- Developed a robust web scraping automation tool to parse and extract structured data (such as movie titles and grid cards) from target web pages.
- Implemented proper HTTP headers, timeout handling, error management (`raise_for_status`), and HTML parsing logic using BeautifulSoup.

Languages

- **Amharic:** Native
- **English:** Professional Working Proficiency